SAWYER WELDEN

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EXPERIENCE

Noregon Systems | Software Engineer Apr 2024 – Present

- Independently built AI-assisted repair-time estimation software with configurable model integrations, .NET APIs, SQL persistence, and human review workflows.
- Modernized data access and expanded regression coverage as part of a six-service .NET migration, preserving existing API contracts.
- Developed C# services for TripVision, a vehicle monitoring platform supporting predictive maintenance across 300,000+ vehicles.
- Designed and implemented billing portal APIs supporting customer management across multiple Noregon products.
- Diagnosed database connection exhaustion with custom load tests and corrected connection lifetimes and transient-failure handling.
- Reduced page-load times by over 40% through SQL stored procedure and Angular optimizations.

Wake Forest School of Medicine | Programmer 3 Jul 2021 – Dec 2023

- Built isolation-forest anomaly detection for the MoTrPAC clinical trial, migrated a SAS pipeline to R, and developed R Shiny quality-control dashboards.
- Co-developed the aorsf R package for survival analysis, including C++ optimizations to Cox regression.
- Trained Python/scikit-learn models for a posture-aware blood-pressure cuff and presented results to healthcare stakeholders.

Rochester Institute of Technology | Graduate Research Assistant Jul 2019 – Dec 2021

- Built Julia and React tools for fair-division simulations, survey analysis, and preference modeling.

TECHNICAL SKILLS

Languages: C#, Python, SQL, Java, TypeScript/JavaScript, R, C++
Backend & data: .NET, ASP.NET Core, SQL Server, Spring WebFlux
Cloud & delivery: Azure, Azure DevOps, CI/CD pipelines, Docker, Git
Web & analytics: Angular, R Shiny, scikit-learn

SELECTED PROJECTS

Piedmont Pauper League | Python, JavaScript · piedmontpauper.org

- Built and maintain a regional trading card game analytics platform with automated tournament-result imports from multiple stores.
- Automated season standings and Elo ranking updates on submission; added player profiles and deck analysis.

Card Game Analytics & Tools · sawyerwelden.com/pauper

- Built a collection of interactive trading card game tools and tournament reports, including deck-similarity maps and clustering, turn-by-turn probability analysis, and visualizations of strategy performance and card usage.

SELECTED PUBLICATIONS · COAUTHOR

Flexible outlier detection in multicenter clinical trials.
Journal of Clinical and Translational Science (2026).

Accelerated and Interpretable Oblique Random Survival Forests.
Journal of Computational and Graphical Statistics (2024).

aorsf: An R package for supervised learning using the oblique random survival forest.
Journal of Open Source Software (2022).

EDUCATION

Rochester Institute of Technology — M.S., Computer Science (2021)
New College of Florida — B.A., Computer Science (2019)